Data Cleaning

[16]:
```python
df['Item Fat Content'] = df['Item Fat Content'].replace({'LF': 'Low Fat', 'low fat' : 'Low Fat', 'reg' : 'Regular'})
```

[21]:
```python
print(df['Item Fat Content'].unique())
```

```
['Regular' 'Low Fat']
```

BUSINESS RE

KPI's REQUIREMENTS

KPI's REQUIREMENTS

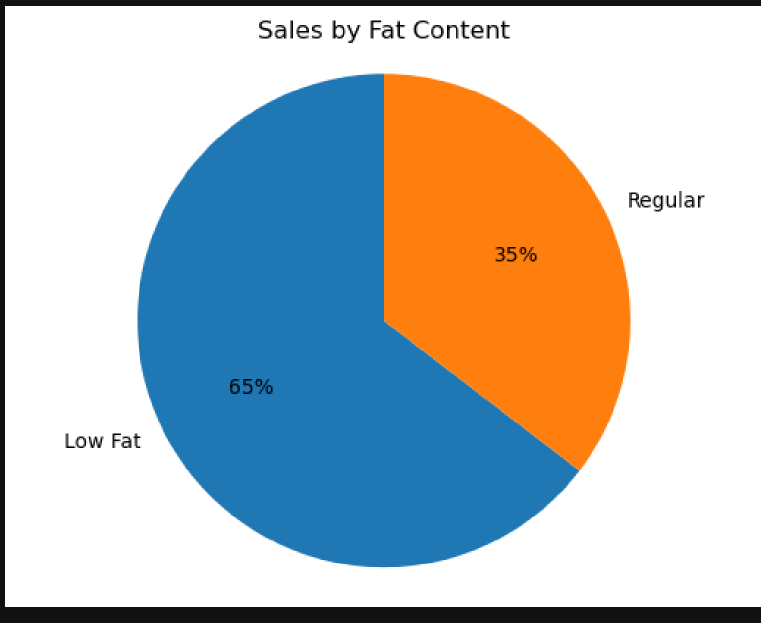
[28]:
```python
#Total Sales
total_sales = df['Sales'].sum()

#Average Sales
avg_sales = df['Sales'].mean()

#No of Items Sold
no_of_items_sold =  df['Sales'].count()

#Average Ratings
avg_ratings = df['Rating'].mean()

#Display

print(f"Total Sales: ${total_sales:,.0f}")
print(f"Average Sales: ${avg_sales:,.1f}")
print(f"No of Items Sold: {no_of_items_sold:,.0f}")
print(f"Average Ratings: {avg_ratings:,.1f}")
```

```
Total Sales: $1,201,681
Average Sales: $141.0
No of Items Sold: 8,523
Average Ratings: 4.0
```

CHARTS REQUIREMENTS

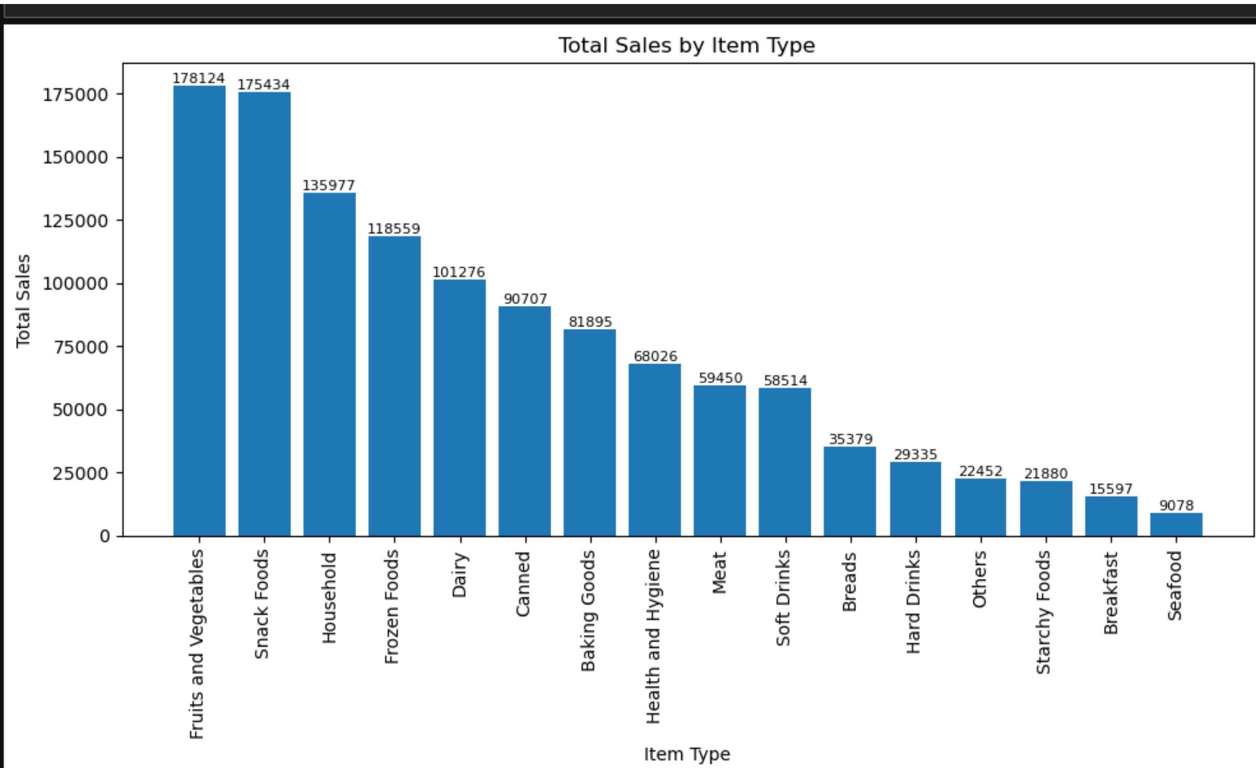
Total Sales by Fat Content

[35]:
```
sales_by_fat = df.groupby('Item Fat Content')['Sales'].sum()

plt.pie(
    sales_by_fat,
    labels=sales_by_fat.index,
    autopct='%.0f%%',
    startangle=90
)

plt.title('Sales by Fat Content')
plt.axis('equal')  # ✅ fixed typo: 'eqaul' → 'equal'
plt.show()
```



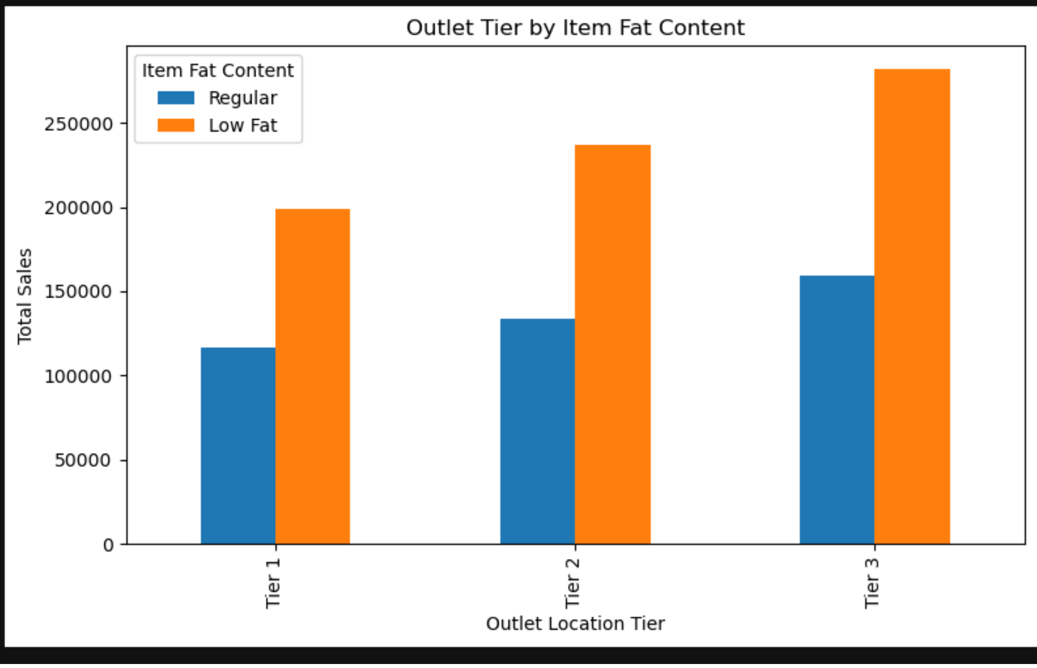
Total Sales by Item Type

[36]:
```
sales_by_type = df.groupby('Item Type')['Sales'].sum().sort_values(ascending=False)

plt.figure(figsize=(10, 6))
bars = plt.bar(sales_by_type.index, sales_by_type.values)

plt.xticks(rotation=90)
plt.xlabel('Item Type')
plt.ylabel('Total Sales')
plt.title('Total Sales by Item Type')

for bar in bars:
    plt.text(
        bar.get_x() + bar.get_width() / 2,
        bar.get_height(),
        f'{bar.get_height():.0f}',
        ha='center',
        va='bottom',
        fontsize=8
    )

plt.tight_layout()
plt.show()
```



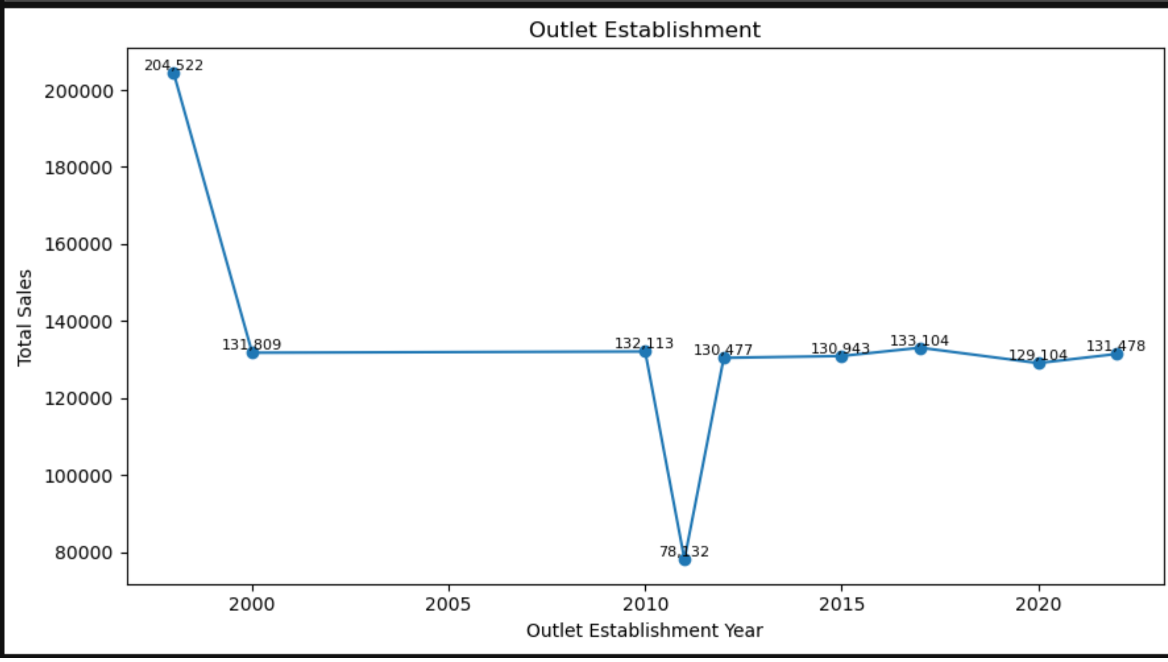
Fat Content by Outlet for Total Sales

[10]:
```
grouped = df.groupby(['Outlet Location Type', 'Item Fat Content'])['Sales'].sum().unstack()
grouped = grouped[['Regular', 'Low Fat']]
ax = grouped.plot(kind='bar', figsize=(8, 5), title='Outlet Tier by Item Fat Content')
plt.xlabel('Outlet Location Tier')
plt.ylabel('Total Sales')
plt.legend(title='Item Fat Content')
plt.tight_layout()
plt.show()
```



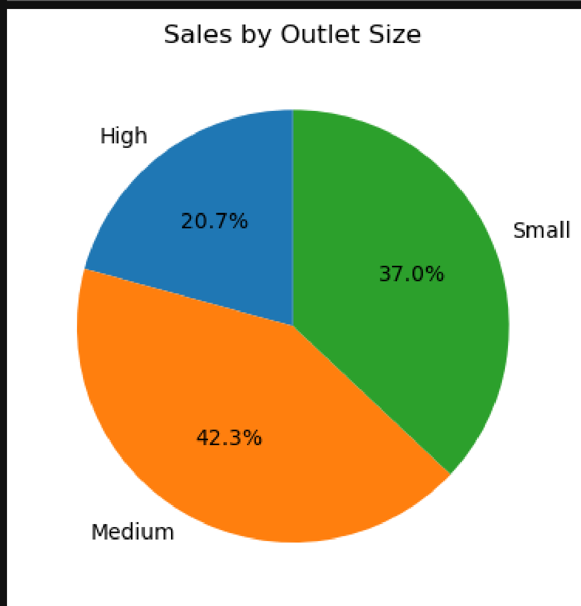
Total Sales by Outlet Establishment

[11]:
```
sales_by_year = df.groupby('Outlet Establishment Year')['Sales'].sum().sort_index()

plt.figure(figsize=(9,5))
plt.plot(sales_by_year.index, sales_by_year.values, marker='o', linestyle='-')

plt.xlabel('Outlet Establishment Year')
plt.ylabel('Total Sales')
plt.title('Outlet Establishment')

for x, y in zip(sales_by_year.index, sales_by_year.values):
    plt.text(x, y, f'{y:,.0f}', ha='center', va='bottom', fontsize=8)

plt.tight_layout()
plt.show()
```



Sales by Outlet Size

[12]:
```
sales_by_size = df.groupby('Outlet Size')['Sales'].sum()

plt.figure(figsize=(4, 4))
plt.pie(sales_by_size, labels=sales_by_size.index, autopct='%1.1f%%', startangle=90)
plt.title('Sales by Outlet Size')
plt.tight_layout()
plt.show()
```



Sales by Outlet Location

[14]:
```
sales_by_location = df.groupby('Outlet Location Type')['Sales'].sum().reset_index()
sales_by_location = sales_by_location.sort_values('Sales', ascending=False)

plt.figure(figsize=(8, 3)) # Smaller height, enough width
ax = sns.barplot(x='Sales', y='Outlet Location Type', data=sales_by_location)

plt.title('Total Sales by Outlet Location Type')
plt.xlabel('Total Sales')
plt.ylabel('Outlet Location Type')

plt.tight_layout() # Ensures layout fits without scroll
plt.show()
```